

Software Engineering – Practice Exam Scenarios

Scenario 1

Hospital Management System for a 500-bed multi-specialty hospital.

Q1: (a) Justify a suitable lifecycle model for this system. (b) Propose and refine major system functions (1.0 level). Q2: (a) List additional essential modules with parameters and purpose. (b) Draw a sequence diagram for a core transaction/use case. Q3: Identify technical, operational, and project risks. Suggest mitigation strategies. Q4: (a) Design 5 detailed test cases for one major function. (b) Perform COCOMO cost estimation assuming appropriate KLOC size and justify model selection.

Scenario 2

Autonomous Car Driving Simulator for testing AI driving algorithms.

Q1: (a) Justify a suitable lifecycle model for this system. (b) Propose and refine major system functions (1.0 level). Q2: (a) List additional essential modules with parameters and purpose. (b) Draw a sequence diagram for a core transaction/use case. Q3: Identify technical, operational, and project risks. Suggest mitigation strategies. Q4: (a) Design 5 detailed test cases for one major function. (b) Perform COCOMO cost estimation assuming appropriate KLOC size and justify model selection.

Scenario 3

Online Banking & Fraud Detection System for a national bank.

Q1: (a) Justify a suitable lifecycle model for this system. (b) Propose and refine major system functions (1.0 level). Q2: (a) List additional essential modules with parameters and purpose. (b) Draw a sequence diagram for a core transaction/use case. Q3: Identify technical, operational, and project risks. Suggest mitigation strategies. Q4: (a) Design 5 detailed test cases for one major function. (b) Perform COCOMO cost estimation assuming appropriate KLOC size and justify model selection.

Scenario 4

Railway Reservation and Dynamic Scheduling System.

Q1: (a) Justify a suitable lifecycle model for this system. (b) Propose and refine major system functions (1.0 level). Q2: (a) List additional essential modules with parameters and purpose. (b) Draw a sequence diagram for a core transaction/use case. Q3: Identify technical, operational, and project risks. Suggest mitigation strategies. Q4: (a) Design 5 detailed test cases for one major function. (b) Perform COCOMO cost estimation assuming appropriate KLOC size and justify model selection.

Scenario 5

Smart City Traffic Control & Monitoring System.

Q1: (a) Justify a suitable lifecycle model for this system. (b) Propose and refine major system functions (1.0 level). Q2: (a) List additional essential modules with parameters and purpose. (b) Draw a sequence diagram for a core transaction/use case. Q3: Identify technical, operational, and project risks. Suggest mitigation strategies. Q4: (a) Design 5 detailed test cases for one major function. (b) Perform COCOMO cost estimation assuming appropriate KLOC size and justify model selection.

Scenario 6

Military Drone Command & Control Platform.

Q1: (a) Justify a suitable lifecycle model for this system. (b) Propose and refine major system functions (1.0 level). Q2: (a) List additional essential modules with parameters and purpose. (b) Draw a sequence diagram for a core transaction/use case. Q3: Identify technical, operational, and project risks. Suggest mitigation strategies. Q4: (a) Design 5 detailed test cases for one major function. (b) Perform COCOMO cost estimation assuming appropriate KLOC size and justify model selection.

Scenario 7

University Learning Management System (LMS).

Q1: (a) Justify a suitable lifecycle model for this system. (b) Propose and refine major system functions (1.0 level). Q2: (a) List additional essential modules with parameters and purpose. (b) Draw a sequence diagram for a core transaction/use case. Q3: Identify technical, operational, and project risks. Suggest mitigation strategies. Q4: (a) Design 5 detailed test cases for one major function. (b) Perform COCOMO cost estimation assuming appropriate KLOC size and justify model selection.

Scenario 8

Air Traffic Control Management System.

Q1: (a) Justify a suitable lifecycle model for this system. (b) Propose and refine major system functions (1.0 level). Q2: (a) List additional essential modules with parameters and purpose. (b) Draw a sequence diagram for a core transaction/use case. Q3: Identify technical, operational, and project risks. Suggest mitigation strategies. Q4: (a) Design 5 detailed test cases for one major function. (b) Perform COCOMO cost estimation assuming appropriate KLOC size and justify model selection.

Scenario 9

Disaster Response Coordination System.

Q1: (a) Justify a suitable lifecycle model for this system. (b) Propose and refine major system functions (1.0 level). Q2: (a) List additional essential modules with parameters and purpose. (b) Draw a sequence diagram for a core transaction/use case. Q3: Identify technical, operational, and project risks. Suggest mitigation strategies. Q4: (a) Design 5 detailed test cases for one major function. (b) Perform COCOMO cost estimation assuming appropriate KLOC size and justify model selection.

Scenario 10

E-commerce Platform with Real-time Inventory and Logistics.

Q1: (a) Justify a suitable lifecycle model for this system. (b) Propose and refine major system functions (1.0 level). Q2: (a) List additional essential modules with parameters and purpose. (b) Draw a sequence diagram for a core transaction/use case. Q3: Identify technical, operational, and project risks. Suggest mitigation strategies. Q4: (a) Design 5 detailed test cases for one major function. (b) Perform COCOMO cost estimation assuming appropriate KLOC size and justify model selection.